

DENNIS FARMER

Ann Arbor, Michigan

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Education

University of Michigan - Ann Arbor, MI

Anticipated May 2027

B.S.E. Data Science - College of Engineering, Minor in Music

GPA: 3.87/4.00

Current Coursework: Web Systems, Regression Analysis, Music and AI, Digital Sound Synthesis

Coursework: Machine Learning, Computational Methods for Statistics, Data Structures and Algorithms,

Probability Theory, Linear Algebra, Discrete Mathematics

Projects

Shazam Music Recognition Clone - Project Lead

June 2025 – Present

Michigan Data Science Team

Ann Arbor, MI

- Replicated the Shazam algorithm in NumPy and integrated it into an Expo app interface, with a web-scraped song library stored using MySQL, in collaboration with one other project co-lead
- Designing and teaching an 8-week curriculum to teach project members the short-time Fourier transform, the Shazam algorithm, SQL, and React Native, for members to create their own version of Shazam

Synthesizer Design

September 2025 – Present

Class Project - Digital Sound Synthesis

Ann Arbor, MI

- Creating a digital synthesizer using the Faust language, incorporating additive, subtractive and frequency modulation (FM) synthesis

Music Recommender System

March 2025 – May 2025

Personal Project

Ann Arbor, MI

- Built an automatic playlist continuation model with a web interface, trained on a web-scraped dataset of 118,000 audio tracks along with user interaction data (Beautiful Soup, scikit-learn, SQL, Streamlit, Plotly)
- Created API to extract audio embeddings using containerized CNN model trained on genre classification (Tensorflow, AWS Lambda, Docker, Flask, ngrok)
- Implemented kernel principle component analysis on audio embeddings, as well as latent matrix factorization methods, for track recommendation via content-based and collaborative filtering respectively

Convolutional Neural Network for Image Classification

September 2023 – December 2023

Michigan Data Science Team

Ann Arbor, MI

- Designed a convolutional neural network architecture to distinguish between AI-generated faces and real human faces with a team of five people (PyTorch)
- Wrote a web application for uploading images and receiving classification results via API, with interpretability via GradCam convolutional layer visualizations (React, Express)

Research Experience

Machine Learning Researcher

May 2025 - Present

Cooperative Institute for Great Lakes Research

Ann Arbor, MI

- Utilizing PyTorch and HPC to recommend ideal buoy sensor placements to enhance weather forecasting
- Training convolutional gaussian neural processes on multi-modal environmental Great Lakes data for statistical downscaling and surface temperature uncertainty quantification (PyTorch)

Roaming Behavior of Domestic Cats

October 2024 - December 2024

Class Research Paper - Computational Methods for Statistics

Ann Arbor, MI

- Developed a beta regression model based on demographic variables to predict proportion of time domestic cats spent away from home, with negligible bias and MSE of 4.052 verified with Monte Carlo simulations
- Concluded cats roam farther away in dry conditions than during rainfall with a linear mixed-effects model, with bootstrapped 95% confidence interval for difference in means of (0.276 km, 1.069 km)

Skills

Languages C++, Python (PyTorch, Pandas, Numpy, Beautiful Soup, Flask), R, SQL, Bash, Faust, ChucK

Tools and Software: Git, Docker, ngrok, Vim, Make, VS Code, GNU Debugger (gdb), Valgrind